



Doing business without advertising is like winking at a girl in the dark. You know what you are doing, but nobody else does.



-Stuart Henderson Britt

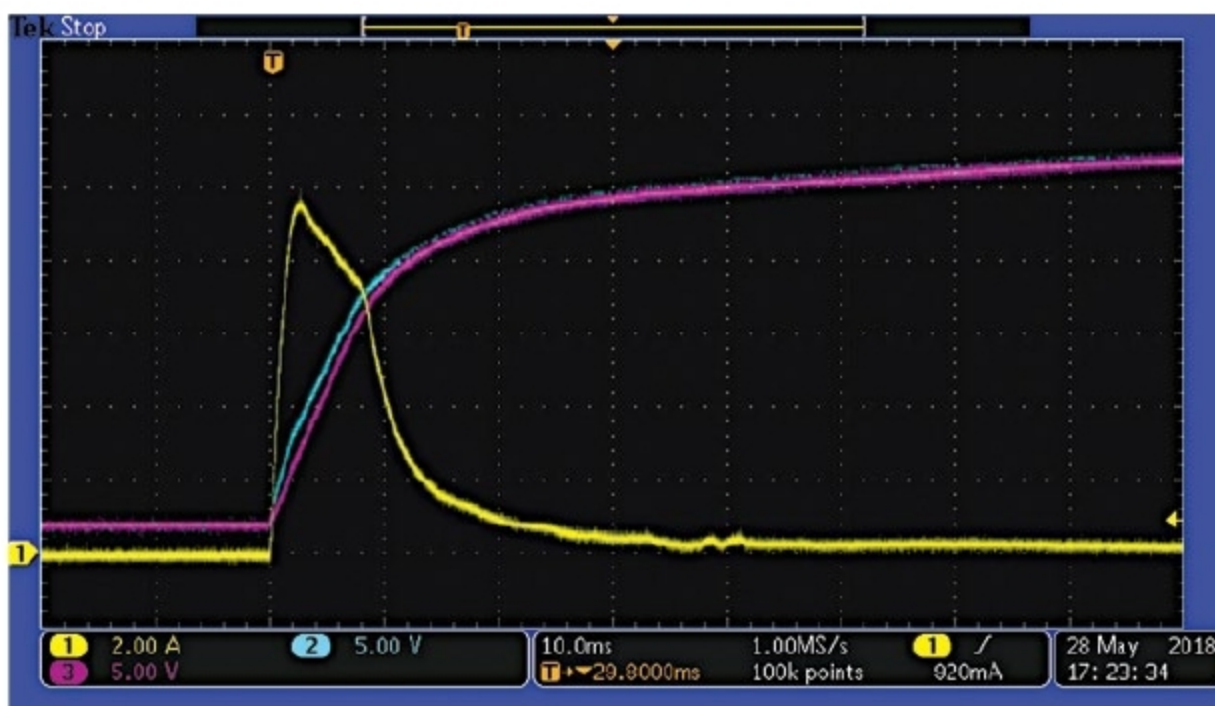


Fig. 3: Inrush voltage of a DC/DC converter in a non-active state (cold-room temperature 25°C)

the lamp (purple line). It is good to see the point at which the device is switched on (marked T, in orange), which reaches its maximum value at around 10A and within 10ms returns to 300mA again.

If you are having issues with inrush voltage affecting a circuit, using a thermistor (NTC) can help.

Electromagnetic compatibility

It is important to establish EMC for the general application. Using a DC/DC converter with an internal filter does not automatically mean that you will adhere to the values specified for the application. This is because EMC compatibility can often be affected by several components.

In many cases, output voltage must be connected with protective earth for safety reasons. This can have a significant impact on EMC. Usually, the power supply manufacturer can offer advice regarding how to adhere to EMC values.

Most power supply manufacturers provide help in the form of suggestions for suitable filters on their websites. For example, at www.tracopower.com these can be downloaded directly from the relevant device's page. If you cannot find the circuit diagrams for the product you have selected, do not hesitate to contact the manufacturer directly on the phone or by email.

Summary

In conclusion, you do not need to put in a huge effort to establish the power

supply components you need for a design if you make use of help from manufacturers and simple resources available.

Before any assessment and selection, it is important to define your requirements clearly. There is a big difference between deciding on what you really need and what you would like.

Coming back to the car manufacturers we referred to at the start of this article, it is important to think about how measurements can be taken correctly, and how every measurement impacts values gathered. Testing strips and laboratory conditions are defined when a circuit is built, but what are the conditions that the circuit will actually be used in?

If there is no suitable device available in the marketplace, maybe, series- or parallel-switching or using a filter might produce the desired results.

Meeting EMC requirements depends heavily on the field of application and its prerequisite conditions, just as with ripple and noise, and inrush current.

It is well-known that there are many manufacturers out there, so the quality as well as price of a specific product can vary greatly.

Is it enough to choose a simple, unbranded, mass-produced device, or will trying to save money come back to bite you further down the line? Do you always need to go for the high-end variant? The final decision is down to you, regardless of any recommendations. **EFY**